# JAVASCRIPT INTERVIEW QUESTIONS

## 1. What is JavaScript?

JavaScript is a high-level, interpreted programming language used to create interactive and dynamic web content. It is a versatile, lightweight, and event-driven language that can run on both the client and server sides.

```
console.log("Hello, JavaScript!");
```

## 2. What are the data types supported by JavaScript?

JavaScript supports the following data types:

- **Primitive Types:** String, Number, BigInt, Boolean, Undefined, Null, Symbol.
- **Non-Primitive Types:** Object (including arrays, functions).

```
let num = 42; // Number
let name = "Siva"; // String
let isStudent = true; // Boolean
let undef; // Undefined
let obj = { key: "value" }; // Object
```

## 3. What is the difference between let, const, and var?

- **var:** Function-scoped, allows redeclaration, hoisted with undefined.
- **let:** Block-scoped, no redeclaration, hoisted but not initialized.
- **const:** Block-scoped, no redeclaration, and must be initialized during declaration.

```
var a = 10;
let b = 20;
const c = 30;
```

## 4. Explain how == and === differ.

- == checks for value equality with type coercion.
- === checks for strict equality without type coercion

```
console.log(5 == "5");  // true
console.log(5 === "5"); // false
```

## 5. What is a closure?

A closure is a function that retains access to its outer scope variables even after the outer function has executed.

```
function outer() {
  let count = 0;
  return function inner() {
    count++;
    return count;
  };
}
const counter = outer();
console.log(counter()); // 1
console.log(counter()); // 2
```

## 6. What is hoisting?

Hoisting is JavaScript's behavior of moving variable and function declarations to the top of their scope during compilation.

```
console.log(a); // undefined (hoisted)
var a = 5;
```

## 7. Explain the concept of this in JavaScript.

this refers to the context in which a function is executed. Its value depends on how the function is called.

```
const obj = {
  name: "Siva",
  greet() {
    console.log(this.name);
  },
};
obj.greet(); // Siva
```

## 8. What are JavaScript prototypes?

Prototypes allow objects to inherit properties and methods from other objects.

```
function Person(name) {
  this.name = name;
}
Person.prototype.greet = function () {
  return `Hello, ${this.name}`;
};
const person = new Person("Siva");
console.log(person.greet()); // Hello, Siva
```

## 9. What is the difference between null and undefined?

- **null:** A deliberate non-value.
- **undefined:** A variable declared but not assigned a value.

```
let x = null;
let y;
console.log(x); // null
console.log(y); // undefined
```

## 10. How does JavaScript handle asynchronous operations?

JavaScript uses the event loop with callbacks, promises, and async/await for asynchronous operations.

### Example: Using a Callback

```
function fetchData(callback) {
  setTimeout(() => {
    callback("Data fetched");
  }, 2000);
}

fetchData((data) => {
  console.log(data); // Logs "Data fetched" after 2 seconds
});
```

## Example: Using a Promise

```
function fetchData() {
  return new Promise((resolve) => {
    setTimeout(() => {
      resolve("Data fetched");
    }, 2000);
  });
}

fetchData().then((data) => {
  console.log(data); // Logs "Data fetched" after 2 seconds
});
```

## Example: Using async/await

```
function fetchData() {
  return new Promise((resolve) => {
    setTimeout(() => {
      resolve("Data fetched");
    }, 2000);
  });
}

async function getData() {
  const data = await fetchData();
  console.log(data); // Logs "Data fetched" after 2 seconds
}

getData();
```

## 11. What is a promise?

A promise represents the eventual completion or failure of an asynchronous operation.

```
const promise = new Promise((resolve, reject) => {
  setTimeout(() => resolve("Success"), 1000);
});
promise.then(console.log); // Success
```

## 12. What are async/await functions?

async/await allows writing asynchronous code that looks synchronous.

```
async function fetchData() {
  const data = await fetch("https://api.example.com");
  return data.json();
}
```

## 13. Explain event delegation in JavaScript.

Event delegation allows you to handle events for multiple child elements at the parent level.

```
document.getElementById("parent").addEventListener("click", (e) => {
  if (e.target.tagName === "BUTTON") {
    console.log("Button clicked!");
  }
});
```

## 14. What are JavaScript modules?

Modules allow you to organize code into reusable files using import and export

```
//Example: module.js
export const greet = () => "Hello, Module!";

//main.js
import { greet } from "./module.js";
console.log(greet());
```

## 15. How can you prevent a function from being called multiple times?

You can use a debounce function.

```
function debounce(func, delay) {
  let timeout;
  return (...args) => {
    clearTimeout(timeout);
    timeout = setTimeout(() => func(...args), delay);
  };
}
```

## 16. What is the event loop?

The event loop processes tasks from the queue and stack for asynchronous operations.

```
console.log("Start");
setTimeout(() => console.log("Timeout"), 0);
console.log("End");
```

## 17. What is the difference between apply() and call() methods?

- **call():** Invokes a function with a specific this value and arguments passed individually.
- **apply():** Similar to call(), but arguments are passed as an array

```
function greet(greeting, punctuation) {
  return `${greeting}, ${this.name}${punctuation}`;
}
const person = { name: "Siva" };

console.log(greet.call(person, "Hello", "!")); // Hello, Siva!
console.log(greet.apply(person, ["Hi", "."])); // Hi, Siva.
```

## 18. What is bind() method used for?

The bind() method creates a new function with a specific this value and optional arguments.

```
const obj = { name: "Siva" };
function greet(greeting) {
  return `${greeting}, ${this.name}`;
}
const boundGreet = greet.bind(obj);
console.log(boundGreet("Hello")); // Hello, Siva
```

## 19. What is a JavaScript event loop?

The event loop continuously checks the call stack and the task queue, executing tasks from the queue when the stack is empty.

```
console.log("Start"); // Executed first, added to the call stack

setTimeout(() => {
  console.log("Timeout callback"); // Added to the task queue, executed later
}, 1000);

console.log("End"); // Executed immediately after "Start"
```

## 20. Explain the concept of "event bubbling" and "event capturing".

- **Event Bubbling:** Events propagate from the target element to the parent elements.
- **Event Capturing:** Events propagate from the parent elements to the target element.

```
document.getElementById("child").addEventListener("click", () =>
console.log("Child"), true); // Capturing
document.getElementById("parent").addEventListener("click", () =>
console.log("Parent"));    // Bubbling
```

## 21. What is the difference between deep copy and shallow copy?

- **Shallow Copy:** Copies only the first layer of an object.
- **Deep Copy:** Copies all layers of an object.

```
let obj = { a: 1, b: { c: 2 } };
let shallow = { ...obj };
let deep = JSON.parse(JSON.stringify(obj));
```

## 22. What are generator functions?

Generators are special functions that can pause execution and resume later.

```
function* generator() {
  yield 1;
  yield 2;
  yield 3;
}
const gen = generator();
console.log(gen.next().value); // 1
console.log(gen.next().value); // 2
```

## 23. What is the new keyword used for?

The new keyword creates an instance of an object from a constructor function.

```
function Person(name) {
  this.name = name;
}
const person = new Person("Siva");
console.log(person.name); // Siva
```

## 24. How do JavaScript's setTimeout and setInterval work?

- **setTimeout:** Executes a function after a specified delay.
- **setInterval:** Repeats execution at specified intervals.

```
setTimeout(() => console.log("Timeout"), 1000);
setInterval(() => console.log("Interval"), 2000);
```

## 25. What is a WeakMap and how is it different from a Map?

- **WeakMap:** Keys are only objects and are garbage collected.
- **Map:** Keys can be any type

```
let obj = { key: "value" };
let weakMap = new WeakMap();
weakMap.set(obj, "data");
```

## 26. What is a Set in JavaScript?

A Set is a collection of unique values.

```
const set = new Set([1, 2, 3, 2]);
console.log(set); // Set { 1, 2, 3 }
```

## 27. What is Object.create() used for?

It creates a new object with a specified prototype.

```
const proto = { greet: () => "Hello" };
const obj = Object.create(proto);
console.log(obj.greet()); // Hello
```

## 28. How does JavaScript's garbage collection work?

JavaScript uses a mark-and-sweep algorithm to identify and remove unused objects.

```
//Example: Garbage Collection in Action
function createObject() {
  const obj = { name: "Siva" }; // Memory is allocated for `obj`
  console.log(obj.name); // "Siva" is accessible here
}

createObject(); // After execution, `obj` is no longer accessible, so it is
garbage-collected.
```

```
//Example: Removing References to Enable Garbage Collection
let obj1 = { name: "Siva" };
let obj2 = obj1; // `obj2` references the same object as `obj1`

console.log(obj1); // Accessible
console.log(obj2); // Accessible

obj1 = null; // The object is still not garbage-collected because `obj2`
references it

obj2 = null; // Now the object is unreachable and will be garbage-collecte
```

## 29. What are "decorators" in JavaScript?

Decorators are functions that modify classes or methods. They are experimental features.

```
function decorator(target) {
  target.isDecorated = true;
}
@decorator
class Example {}
console.log(Example.isDecorated); // true
```

## 30. Explain the difference between prototype and __proto__.

- **prototype:** An object associated with functions for inheritance.
- **__proto__:** A reference to the object's prototype.

```
function Person() {}
const person = new Person();
console.log(person.__proto__ === Person.prototype); // true
```